Eigen-Séta: AVX-512 IFMA field multiplication

aburan28/eigenx#6. Measured on a shared 4-vCPU Xeon @ 2.10 GHz VM; parameters, outputs and security estimate unchanged.

Decapsulation wall time (medians, measured)

set	main 580b633	IFMA, default build	IFMA + native
L1 (n=15)	1.050 s	0.635 s (1.65×)	0.656 s (1.60×)
L3 (n=6)	4.989 s	1.605 s (3.11×)	1.413 s (3.53×)
L5 (n=3)	11.07 s	3.345 s (3.31×)	2.954 s (3.75×)

Run run-20261018T0320Z-ifma (director-compiled card, executor-run). Encapsulation main → IFMA: 0.165 → 0.109 s (L1), 1.024 → 0.310 s (L3), 2.498 → 0.880 s (L5). Versus the white-paper run (d89d790): L1 1.9×, L5 3.8× (derived).

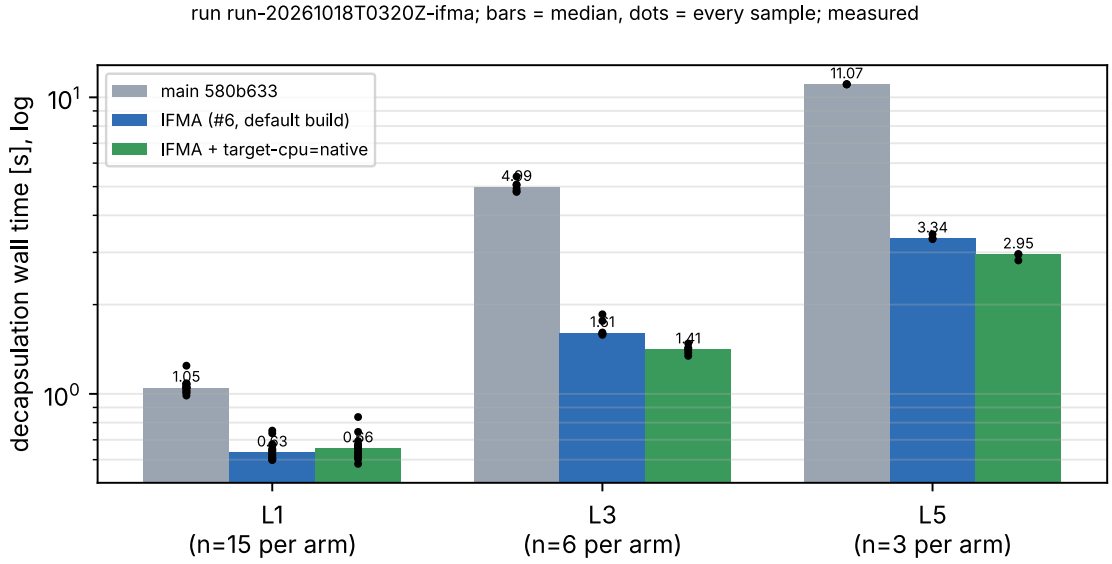

Figure 1: Bars median, dots every sample, log scale; measured.

Constant time (measured)

dudect on L1 decapsulation, 1,000 samples per class: IFMA max |t| 0.87, portable 2.14 (threshold 4.5). Every branch of the IFMA kernels compares counters or lengths only. Results byte-identical to the portable multiplier on all five sets. A field-level fixed-vs-random dudect flags both paths (including the instruction-count-checked portable one) and is not used as evidence.

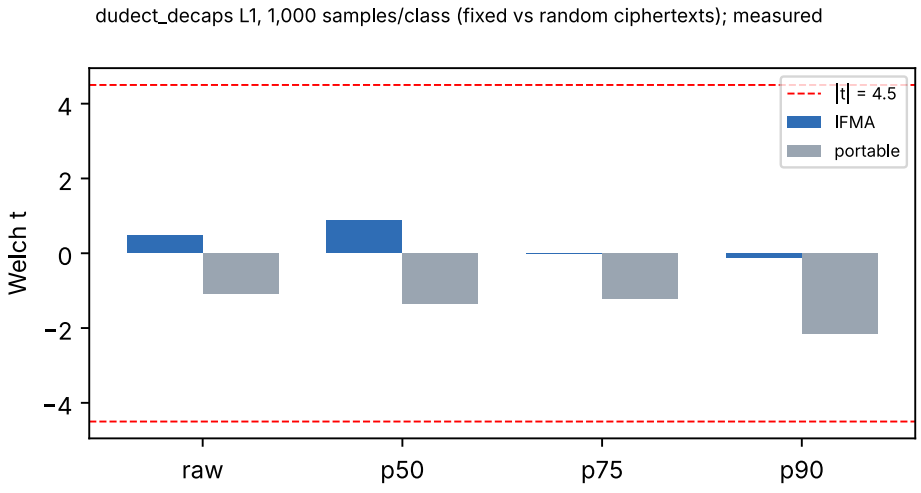

Figure 2: Welch t per crop; measured.

Distance to practicality

L1 decapsulation 0.635 s; a practical PKE needs tens of milliseconds. Next: the theta chain step (618–670 F_{p^2} mults measured vs about 256 modelled), F_p add/sub (about 16%), the public 3^{F3} basis (about 9%), multi-core chains.